

Report R24 — Does Epigenetic Modulation Reverse the Drug-Resistance Programme?

SMARCA4 knockdown in osimertinib-resistant lung cancer cells. The knockdown worked. The reversal was partial in one direction only, and it did not replicate.

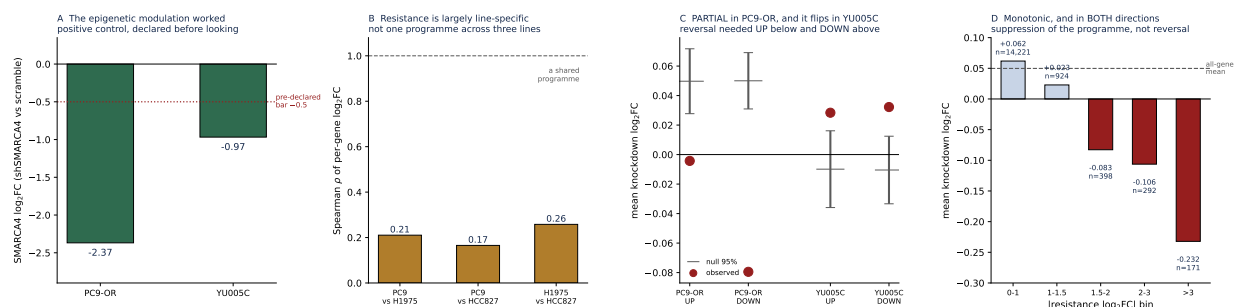
20 August 2026 • Quantara • Peer-review audit copy

Summary

SMARCA4 (BRG1) is the ATPase of the SWI/SNF chromatin remodelling complex, so knocking it down in an osimertinib-resistant cell line *is* epigenetic modulation of a resistant state. That makes “reprogramming drug resistance through epigenetic modulation” a testable claim rather than an assertion, and GSE202859 contains the experiment: three parental/resistant pairs plus SMARCA4 shRNA knockdown in two resistant models.

The analysis plan was SHA-256 hashed at `bffbaad89936c522...` after reading only the TPM column headers and *before any expression value was compared between any two groups*.

Verdict: PARTIAL_one_direction_only, by the plan’s own rule. Reversal required resistance-UP genes to fall below the null *and* resistance-DOWN genes to rise above it. The first happened; the second did not. In the second model the whole pattern inverts.



A Knockdown verification. **B** Cross-line consistency of the resistance programme. **C** The primary test in both models. **D** Knockdown response by magnitude of resistance change.

1 What was measured

The knockdown worked, comfortably. SMARCA4 log₂FC **−2.37** in PC9-OR and **−0.97** in YU005C, against a pre-declared bar of **−0.5**. This was a gate: had it failed, the plan required halting without interpreting anything downstream.

Resistance changes the transcriptome. 1,785, 1,480 and 2,367 genes at $|\log_2 \text{FC}| > 1$ in PC9, H1975 and HCC827 respectively, from 16,006 retained genes.

But the resistance programme is largely line-specific. Pairwise Spearman correlation of per-gene log₂FC between the three lines is only **0.17–0.26**. A programme shared across EGFR-mutant lines would sit near 1. This bears directly on the premise that there is *a* resistance programme to reprogramme.

1.1 The primary test

Resistance sets were defined from the PC9 parental pair alone — 789 up, 996 down — so the sets never touch a knockdown sample. Knockdown log₂FC was then compared against 2,000 random gene sets of matched size.

Model / set	n	mean kd $\log_2\text{FC}$	null 95%	reading
PC9-OR, resistance UP	789	-0.0043	[+0.028, +0.072]	below null — <i>reversal</i>
PC9-OR, resistance DOWN	996	-0.0795	[+0.031, +0.069]	below null — <i>not reversal</i>
YU005C, resistance UP	789	+0.0284	[-0.036, +0.016]	above null — opposite
YU005C, resistance DOWN	996	+0.0322	[-0.033, +0.012]	above null — opposite

All four are separable from their nulls at $p \leq 0.004$. What they are not is a reversal: knockdown pushes resistance-associated genes *down in both directions* in PC9-OR, and up in both directions in YU005C.

1.2 One clean finding

Binning genes by how much they changed on becoming resistant:

resistance $\log_2\text{FC}$	n	mean knockdown $\log_2\text{FC}$
0-1	14,221	+0.062
1-1.5	924	+0.023
1.5-2	398	-0.083
2-3	292	-0.106
> 3	171	- 0.232

Monotonic across five bins: the more a gene changed on becoming resistant, the more SMARCA4 knockdown lowers it. That is a real dose-dependent relationship and it is the most interesting number in the report — but it operates on resistance-associated genes regardless of *which way* they moved, which is suppression of the programme rather than reversal of it.

2 Why the honest reading is weak

Effect sizes are minute. -0.004 to -0.079 $\log_2\text{FC}$. Statistically separable from a 2,000-permutation null, biologically close to nothing.

It does not replicate. YU005C inverts every sign. One model showing partial reversal and a second showing the opposite is not a finding about SMARCA4; it is two observations that disagree.

The premise is shakier than the result. With cross-line consistency at 0.17-0.26, “the resistance programme” is not one thing across EGFR-mutant lines, so a therapy that reprogrammes *it* has an ill-defined target.

3 Limitations

Two to three replicates per group. This is a small experiment and no negative-binomial count model was used — TPMs were log-transformed and compared by group means, as the plan specified.

shRNA has off-target effects. SMARCA4-specific attribution is not established by this design; two independent hairpins agreeing is supportive, not conclusive.

The resistance set and the knockdown share a lineage. Sets were defined in PC9 and applied to knockdown in PC9-OR, so H2 is internally consistent but not independent. YU005C is the independent test, and it failed.

Transcriptional reversal is not therapy. No viability, growth or IC_{50} endpoint is analysed here. The plan’s refusals forbid claiming a therapeutic effect and this report does not.

Cell lines are not patients.

Hash-before-outcome only. Plan author and executor were the same person — the weaker form of the protocol, not the structural isolation used in R16.

Reproducing this report

evidence/PREREG_PLAN_R24.yaml (hashed bffbaad89936c522...), evidence/PREREGISTRATION_R24.json, evidence/run_r24.py, evidence/results.json, evidence/gates.json (8 gates), evidence/dose_trend.json, evidence/kd_lfc_PC90R.npy, evidence/res_lfc_PC9.npy, evidence/genes.txt. Figure: python3 make_figure.py. Source data: GEO GSE202859.